

Using the universal property of $\mathrm{THH}(k)$ as a commutative algebra, we find that $k \otimes_{k \otimes \mathrm{THH}(k)} k \simeq \mathrm{THH}(k) \otimes (k \otimes_{\mathrm{THH}(k)} k)$, and using now Bökstedt's equivalence $\mathrm{THH}(k) \simeq k[\Omega S^3]$, we find $k \otimes_{\mathrm{THH}(k)} k \simeq k[S^3]$, so that, in total,

$$\mathrm{map}_{k \otimes \mathrm{THH}(k)}(k, k) \simeq \prod_n (\Omega^{2n} k \oplus \Omega^{2n+3} k)$$

To get k -linear endomorphisms of HH_k , one simply adjoins $[S^1]$ to this. $\triangleleft$

Remark 3.4. In the previous examples, it is not clear to the author what the extra operations “do”. Similarly to the situation in Remark 2.15, we do not know how to “name” them, and it would probably be worthwhile to spend time figuring out what these operations do, or are. $\triangleleft$

4 Operations on THH of $\mathcal{O}$ -algebras

The goal of this section is to initiate the study of endomorphisms of THH viewed as a functor $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Cat}^{\mathrm{perf}}) \rightarrow \mathrm{Sp}$, that is, to study the extra operations that arise on $\mathrm{THH}(C)$ when C has a particular kind of multiplicative structure encoded by a one-colored ∞ -operad $\mathcal{O}$. Our main result in this section is Theorem B.

As is clear from the statement of Theorem B, the case of endomorphisms of THH viewed as a functor on $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Cat}^{\mathrm{perf}})$ is more subtle.

We first describe the general approach to this question, and then specialize to get the precise results that we claimed.

We let $U : \mathrm{Alg}_{\mathcal{O}}(\mathrm{Cat}^{\mathrm{perf}}) \rightarrow \mathrm{Cat}^{\mathrm{perf}}$ denote the forgetful functor, with a left adjoint F such that $UF \simeq \bigoplus_n (\mathcal{O}(n) \otimes (-)^{\otimes n})_{h\Sigma_n}$ [Lur12, Proposition 3.1.3.13].

The idea now is that by general adjunction nonsense,

$$\mathrm{Map}_{\mathrm{Fun}(\mathrm{Alg}_{\mathcal{O}}(\mathrm{Cat}^{\mathrm{perf}}), \mathrm{Sp})}(\mathrm{THH} \circ U, \mathrm{THH} \circ U) \simeq \mathrm{Map}_{\mathrm{Fun}(\mathrm{Cat}^{\mathrm{perf}}, \mathrm{Sp})}(\mathrm{THH}, \mathrm{THH} \circ UF)$$

and so we are “left with” understanding $\mathrm{THH} \circ UF$.

For this, we compute each of the individual terms of

$$\mathrm{THH} \circ UF = \bigoplus_n \mathrm{THH}((\mathcal{O}(n) \otimes (-)^{\otimes n})_{h\Sigma_n})$$

Proposition 4.1. *Let $\mathcal{O}$ be a space with a Σ_n -action. There is an equivalence, natural in $C \in \mathrm{Cat}^{\mathrm{perf}}$:*

$$\mathrm{THH}((\mathcal{O} \otimes C^{\otimes n})_{h\Sigma_n}) \simeq \bigoplus_{\sigma \in \Sigma_n / \mathrm{conj}} (\mathcal{O}^\sigma \otimes \mathrm{THH}(C)^{\otimes n(\sigma)})_{hC(\sigma)}$$

where for $\sigma \in \Sigma_n$, $n(\sigma)$ is the number of cycles appearing in σ , $C(\sigma)$ is the centralizer of σ in Σ_n and $\mathcal{O}^\sigma = L(\mathcal{O}_{h\Sigma_n}) \times_{LB\Sigma_n} \{\sigma\}$ with its residual $C(\sigma)$ -action¹³.

¹³Here, L denotes the free loop space, i.e. $\mathrm{Map}(S^1, -)$.